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Patent Public Search | Text View

United States Patent Application Publication

20250274680

Kind Code

A1

Publication Date

August 28, 2025

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Image device capable of switching between global shutter mode and dynamic vision sensor mode

Abstract

An image device has a pixel array and a controller. The pixel array includes a plurality of sensor pixels. The controller is coupled to the pixel array to switch the image device between a first mode and a second mode. When the image device is in the first mode, the controller manipulates the sensor pixels according to a first timing scheme, configuring the image device to function as a global shutter image sensor for capturing images of a scene. When the image device is in the second mode, the controller manipulates the sensor pixels according to a second timing scheme, configuring the image device to function as a dynamic vision sensor (DVS) for detecting changes in the scene.

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Appl. No.: 18/587903

Filed: February 26, 2024

Publication Classification

Int. Cl.: H04N25/77 (20230101); H04N25/78 (20230101)

U.S. Cl.:

CPC H04N25/77 (20230101); H04N25/78 (20230101);

Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention is related to an image device, and more particularly to an image device switching between a global shutter mode and a dynamic vision sensor (DVS) mode.

2. Description of the Prior Art

[0002] A global shutter image device is a camera or other image sensor that exposes all of its pixels simultaneously. This allows it to capture images of moving objects without any distortion or blur. Global shutter devices are often used in applications where it is important to accurately capture motion, such as in industrial automation, surveillance, and medical imaging.

[0003] A dynamic vision sensor (DVS) is a type of camera that captures images by detecting changes in light intensity over time. This allows it to capture images of moving objects without any distortion or blur, even at high frame rates. DVS devices are often used in applications where it is important to accurately capture motion, such as in robotics, autonomous vehicles, and machine vision.

[0004] However, in the prior art, there was no single image device that could be used as both a global shutter image device and a dynamic vision sensor (DVS).

SUMMARY OF THE INVENTION

[0005] An embodiment of the present invention provides an image device having a pixel array and a controller. The pixel array comprises a plurality of sensor pixels. The controller is coupled to the pixel array and configured to switch the image device between a first mode and a second mode. When the image device is in the first mode, the controller manipulates the sensor pixels according to a first timing scheme, configuring the image device to function as a global shutter image sensor for capturing images of a scene. When the image device is in the second mode, the controller manipulates the sensor pixels according to a second timing scheme, configuring the image device to function as a dynamic vision sensor (DVS) for detecting changes in the scene.

[0006] These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a functional block diagram of an image device according to an embodiment of the present invention.

[0008] FIG. 2 is a circuit diagram of a sensor pixel of the image device shown in FIG. 1.

[0009] FIG. 3 is a diagram illustrating a first timing scheme of the image device shown in FIG. 1.

[0010] FIG. 4 is a diagram illustrating a second timing scheme of the image device shown in FIG. 1.

DETAILED DESCRIPTION

[0011] Please refer to FIG. 1. FIG. 1 is a functional block diagram of an image device 1 according to an embodiment of the present invention. The image device 1 comprises a controller 10 and a pixel array 12. The pixel array 12 comprises a plurality of sensor pixels 20. The sensor pixels 20 are arranged in M rows and N columns. M and N are integers greater than 1. The controller 10 is coupled to the pixel array 12 and is configured to switch the image device 1 between a first mode and a second mode. When the image device 1 is in the first mode, the controller 10 manipulates the sensor pixels 20 according to a first timing scheme, configuring the image device 1 to function as a global shutter image sensor for capturing images of a scene. When the image device 1 is in the second mode, the controller 10 manipulates the sensor pixels 20 according to a second timing

scheme, configuring the image device **1** to function as a dynamic vision sensor (DVS) for detecting changes in the scene. In addition, the image device **1** may further comprise a lens or lenses to focus light from the scene onto the sensor pixels **20** of the pixel array **12**. The sensor pixels **20** convert the received light into an image IMG1 of the scene in the first mode or a dynamic vision image IMG2 in the second mode.

[0012] Please refer to FIG. 2. FIG. 2 is a circuit diagram of one of the sensor pixels **20** of the image device **1** shown in FIG. 1. Each of the sensor pixels **20** comprises a photodiode PD, a first switch Q1, a second switch Q2, a reset switch Q3, and a readout circuit 22. The photodiode PD is configured to detect light. The first switch Q1 is coupled between a first power supply terminal VDD and a first end N1 of the photodiode PD, and is configured to reset a voltage level of the first end N1 of the photodiode PD. The first power supply terminal VDD provides a positive reference voltage. The second switch Q2 is coupled between the first end N1 of the photodiode PD and a floating diffusion node Sn, and is configured to transfer the voltage level of the first end N1 of the photodiode PD to the floating diffusion node Sn. A parasitic capacitance Cs exists between the floating diffusion node Sn and a second power supply terminal VSS. The second power supply terminal VSS may provide a ground voltage. The reset switch Q3 is coupled between the first power supply terminal VDD and the floating diffusion node Sn, and is configured to reset a voltage level of the floating diffusion node Sn. The readout circuit 22 is coupled to the second switch Q2 and the reset switch Q3, and is configured to output a readout signal Sout based on the voltage level of the floating diffusion node Sn. The readout circuit 22 comprises a transistor Q4 and a readout switch Q5. A first end of the transistor Q4 is coupled to the first power supply terminal VDD, a second end of the transistor Q4 is coupled to the readout switch Q5, and a control end of the transistor Q4 is coupled to the floating diffusion node Sn. The readout switch Q5 is configured to control electrical connection between the second end of the transistor Q4 and an output end Nout of the readout circuit 22. The readout signal Sout is outputted via the output end Nout. In addition, the second end of the photodiode PD is coupled to the second power supply terminal VSS.

[0013] In the embodiment, the first switch Q1, the second switch Q2, the reset switch Q3, the transistor Q4, and the readout switch Q5 are N-type transistors, and the sensor pixel **20** is a 5T pixel (i.e., a pixel having five transistors). The gates of the first switch Q1, the second switch Q2, the reset switch Q3, and the readout switch Q5 respectively receive a first control signal S1, a second control signal S2, a reset signal Sr, and a readout control signal RSL. The gate of the transistor Q4 is coupled to the floating diffusion node Sn. When the voltage level of the floating diffusion node Sn is high, the transistor Q4 is turned on. When the voltage level of the floating diffusion node Sn is low, the transistor Q4 is turned off.

[0014] Please refer to FIGS. 1 to 3. FIG. 3 is a diagram illustrating a first timing scheme of the image device **1** shown in FIG. 1. When the image device **1** is in the first mode, the controller **10** manipulates the sensor pixels **20** according to the first timing scheme, configuring the image device **1** to function as a global shutter image sensor for capturing images of a scene. The image device **1** generates an image IMG1 of the scene every frame period T1. As shown in FIG. 3, the frame period T1 is equal to the duration between two adjacent rising edges of the first control signal S1, which occur at time points t1 and t4.

[0015] During a reset period P1, the first control signal S1 and the reset signal Sr are high, the readout control signal RSL and the second control signal S2 are low. This causes the first switch Q1 and the reset switch Q3 to turn on, and the second switch Q2 and the readout switch Q5 to turn off. Accordingly, the voltage level of the first end N1 of the photodiode PD is reset to be equal to the first reference voltage provided by the first power supply terminal VDD. Because the image device **1** functions as a global shutter image sensor when the image device **1** is in the first mode, the first switches Q1 and the reset switches Q3 of all of the sensor pixels **20** of the pixel array **12** are turned on during the reset period P1. Accordingly, the voltage levels of the first ends N1 of the photodiodes PD of all of the sensor pixels **20** and the voltage levels of floating diffusion nodes Sn

of all of the sensor pixels **20** are reset to be equal to the first reference voltage provided by the first power supply terminal VDD.

[0016] At time point **t2**, the first control signal **S1** is pulled from high to low, turning off the first switch **Q1**, stopping resetting the voltage level of the first end **N1**, and enabling the photodiode **PD** to start to accumulate charges at the first end **N1** of the photodiode **PD**.

[0017] During a sensing period **P2** after the reset period **P1**, the first control signal **S1**, the readout control signal **RSL**, the reset signal **Sr**, and the second control signal **S2** are low. This causes the first switch **Q1**, the second switch **Q2**, the reset switch **Q3**, and the readout switch **Q5** to turn off so as to accumulate charges at the first end **N1** of the photodiode **PD**. Because the image device **1** functions as a global shutter image sensor when the image device **1** is in the first mode, the first switches **Q1**, the second switches **Q2**, the reset switches **Q3**, and the readout switches **Q5** of all of the sensor pixels **20** of the pixel array **12** are turned off during the sensing period **P2**. Accordingly, charges are accumulated at the first ends **N1** of the photodiodes **PD** of all of the sensor pixels **20** of the pixel array **12**, causing the voltage levels of the first ends **N1** of the photodiodes **PD** of all of the sensor pixels **20** to be changed by the accumulated charges.

[0018] During a transferring period **P3** after the sensing period **P2**, the first control signal **S1**, the readout control signal **RSL**, and the reset signal **Sr** are low, and the second control signal **S2** is high. This causes the first switch **Q1**, the reset switch **Q3**, and the readout switch **Q5** to turn off, and the second switch **Q2** to turn on, so as to transfer the voltage level of the first end **N1** of the photodiode **PD** to the floating diffusion node **Sn**. Because the image device **1** functions as a global shutter image sensor when the image device **1** is in the first mode, the first switches **Q1**, the reset switches **Q3**, and the readout switches **Q5** of all of the sensor pixels **20** of the pixel array **12** are turned off, and the second switches **Q2** of all of the sensor pixels **20** of the pixel array **12** are turned on during the transferring period **P3**. Accordingly, the voltage levels of the first ends **N1** of the photodiodes **PD** of all of the sensor pixels **20** are simultaneously transferred to corresponding floating diffusion nodes **Sn** of all of the sensor pixels **20** during the transferring period **P3**.

[0019] At time point **t3**, the second control signal **S2** is pulled from high to low, turning off the second switch **Q2**, and locking the voltage level of the floating diffusion node **Sn** to a current voltage level of the first end **N1**. Because the photodiode **PD** starts to accumulate charges at the first end **N1** of the photodiode **PD** at time point **t2**, and the voltage level of the floating diffusion node **Sn** is locked at time point **t3**, the locked voltage level of the floating diffusion node **Sn** is determined by the total charges accumulated at the first end **N1** of the photodiode **PD** between time points **t2** and **t3**. When the image device **1** is in the first mode, the period between **t2** and **t3**, as labeled **P01**, is the exposure time and directly affects the final voltage level at the floating diffusion node **Sn**.

[0020] During a readout period **P4** after the transferring period **P3**, the first control signal **S1**, the reset signal **Sr**, and the second control signal **S2** are low, and the readout control signal **RSL** is high. This causes the first switch **Q1**, the reset switch **Q3**, and second switch **Q2** to turn off, and the readout switch **Q5** to turn on, so as to enable the readout circuit **22** to output the readout signal **Sout** based on the voltage level of the floating diffusion node **Sn**. Even though the image device **1** functions as a global shutter image sensor when the image device **1** is in the first mode, the sensor pixels **20** of the image device **1** can be read out non-simultaneously, either row by row or column by column, instead of being read out all at once. For example, during the readout period **P4**, the sensor pixels **20** of the image device **1** may be read line by line in a time-division manner. During the readout period **P4**, the first switches **Q1**, the reset switches **Q3**, and second switches **Q2** of the sensor pixels **20** arranged in the same line of the pixel array **12** are turned off, and the readout switches **Q5** arranged in the same line of the pixel array **12** are turned on, to enable the readout circuits **22** of the sensor pixels **20** arranged in the same line to output readout signals **Sout** based on the voltage levels of the floating diffusion nodes **Sn** of the sensor pixels **20** arranged in the same line of the pixel array **12**. The same line of the pixel array **12** may be one of the **M** rows or one of

the N columns of the sensor pixels **20**.

[0021] During a reset period **P5** after the readout period **P4**, the first control signal **S1** and the second control signal **S2** are low, and the readout control signal **RSL** and the reset signal **Sr** are high. This causes the first switch **Q1** and second switch **Q2** to turn off, and the reset switch **Q3** and the readout switch **Q5** to turn on, so as to reset the voltage level of the floating diffusion node **Sn** to a corresponding reference level. Because the reset switch **Q3** is turned on, the corresponding reference level is approximately equal to the first reference voltage provided by the first power supply terminal **VDD**. During the reset period **P5**, the voltage levels of the floating diffusion nodes **Sn** of the sensor pixels **20** may be reset line by line in a time-division manner. In detail, during the reset period **P5**, the first switches **Q1** and second switches **Q2** of the sensor pixels **20** arranged in the same line of the pixel array **12** are turned off, and the reset switches **Q3** and the readout switches **Q5** arranged in the same line of the pixel array **12** are turned on, to reset the voltage levels of the floating diffusion nodes **Sn** of the sensor pixels **20** arranged in the same line to corresponding reference levels, which may vary due to manufacturing variations in the sensor pixels **20**.

[0022] During a readout period **P6** after the reset period **P5**, the first control signal **S1**, the reset signal **Sr**, and the second control signal **S2** are low, and the readout control signal **RSL** is high. This causes the first switch **Q1**, the reset switch **Q3**, and second switch **Q2** to turn off, and the readout switch **Q5** to turn on, so as to enable the readout circuit **22** to output the readout signal **Sout** based on the voltage level of the floating diffusion node **Sn**. During the readout period **P6**, the sensor pixels **20** of the image device **1** may be read line by line in a time-division manner. In detail, during the readout period **P6**, the first switches **Q1**, the reset switches **Q3**, and second switches **Q2** of the sensor pixels **20** arranged in the same line of the pixel array **12** are turned off, and the readout switches **Q5** arranged in the same line of the pixel array **12** are turned on, to enable the readout circuits **22** of the sensor pixels **20** arranged in the same line to output readout signals **Sout** based on the voltage levels of the floating diffusion nodes **Sn** of the sensor pixels **20** arranged in the same line of the pixel array **12**.

[0023] The controller **10** manipulates the sensor pixels **20** to perform correlated double sampling (CDS) operations based on the first timing scheme as illustrated in FIG. 3. Specifically, CDS operations involve comparing the readout signal **Sout** of each sensor pixel **20** during the readout period **P4** with the same signal during the readout period **P6**. CDS is a method to measure electrical values such as voltages or currents that allows removing an undesired offset. It is often used when measuring sensor outputs. The output of the sensor is measured twice: once in a known condition and once in an unknown condition. The value measured from the known condition is then subtracted from the unknown condition to generate a value with a known relation to the physical quantity being measured. the CDS operations use a noise reduction technique in which a reference voltage of the sensor pixel **20** (i.e., the voltage level of the readout signal **Sout** within the readout period **P6**) is subtracted from the signal voltage of the sensor pixel **20** (i.e., the voltage level of the readout signal **Sout** within the readout period **P4**), to cancel thermal noise (i.e., Johnson-Nyquist noise or kTC noise). Assuming that the voltage level of the readout signal **Sout** of a sensor pixel **20**, located in the m.sup.th row and n.sup.th column of the pixel array **12**, read within the readout period **P4** is **V1** (m, n), and the voltage level of the readout signal **Sout** of the same sensor pixel **20** read within the readout period **P6** is **V2** (m, n), a difference **D1** (m, n) between **V1** (m, n) and **V2** (m, n) can be represented by the following equation (1):

[00001]
$$D1(m, n) = V1(m, n) - V2(m, n) \quad (1)$$

[0024] Where $1 \leq m \leq M$, $1 \leq n \leq N$, **M** is the total number of rows of the sensor pixels **20** in the pixel array **12**; and **N** is the total number of columns of the sensor pixels **20** in the pixel array **12**.

[0025] The CDS operations are performed according to the difference **D1** (m, n) of each sensor pixel **20**, and the image device **1** in the first mode generates one image **IMG1** of the scene based on

the difference $D1(m, n)$ of each sensor pixel **20** in each frame period **T1**.

[0026] Please refer to FIGS. **1**, **2**, and **4**. FIG. **4** is a diagram illustrating a second timing scheme of the image device **1** shown in FIG. **1**. When the image device **1** is in the second mode, the controller **10** manipulates the sensor pixels **20** according to the second timing scheme, configuring the image device **1** to function as a dynamic vision sensor (DVS) for detecting changes in the scene. The image device **1** in the second mode generates a dynamic vision image IMG2 every frame period **T2**. As shown in FIG. **4**, the frame period **T2** is equal to the duration between two adjacent rising edges of the first control signal **S1**, which occur at time points t_a and t_d .

[0027] During a reset period **R1**, the first control signal **S1** is high, the readout control signal **RSL**, the reset signal **Sr**, and the second control signal **S2** are low. This causes the first switch **Q1** to turn on, and the second switch **Q2**, the reset switch **Q3**, and the readout switch **Q5** to turn off.

Accordingly, the voltage level of the first end **N1** of the photodiode **PD** is reset to be equal to the first reference voltage provided by the first power supply terminal **VDD**. During the reset period **R1**, the sensor pixels **20** of the image device **1** may be reset line by line in a time-division manner. In detail, during the reset period **R1**, the first switches **Q1** of the sensor pixels **20** arranged in the same line of the pixel array **12** are turned on, and the second switches **Q2**, the reset switches **Q3**, and the readout switches **Q5** of the sensor pixels **20** arranged in the same line of the pixel array **12** are turned off. Accordingly, the voltage levels of the first ends **N1** of the photodiodes **PD** of the sensor pixels **20** arranged in the same line of the pixel array **12** are reset within the reset period **R1**.

[0028] At time point t_b , the first control signal **S1** is pulled from high to low, turning off the first switch **Q1**, stopping resetting the voltage level of the first end **N1**, and enabling the photodiode **PD** to start to accumulate charges at the first end **N1** of the photodiode **PD**.

[0029] During a sensing period **R2** after the reset period **R1**, the first control signal **S1**, the reset signal **Sr**, and the second control signal **S2** are low, and the readout control signal **RSL** is high. This causes the first switch **Q1**, the reset switch **Q3**, and second switch **Q2** to turn off, and the readout switch **Q5** to turn on, so as to accumulate charges at the first end **N1** of the photodiode **PD** and enable the readout circuit **22** to output the readout signal **Sout** based on the voltage level of the floating diffusion node **Sn**. During the sensing period **R2**, the sensor pixels **20** of the image device **1** may be read line by line in a time-division manner. In detail, during the sensing period **R2**, the first switches **Q1**, the reset switches **Q3**, and second switches **Q2** of the sensor pixels **20** arranged in the same line of the pixel array **12** are turned off, and the readout switches **Q5** arranged in the same line of the pixel array **12** are turned on, to accumulate charges at the first ends **N1** of the photodiodes **PD** of the sensor pixels **20** arranged in the same line of the pixel array **12** and enable the readout circuits **22** of the sensor pixels **20** arranged in the same line to output readout signals **Sout** based on the voltage levels of the floating diffusion nodes **Sn** of the sensor pixels **20** arranged in the same line of the pixel array **12**.

[0030] During a reset period **R3** after the sensing period **R2**, the first control signal **S1** and the second control signal **S2** are low, and the readout control signal **RSL** and the reset signal **Sr** are high. This causes the first switch **Q1** and second switch **Q2** to turn off, and the reset switch **Q3** and the readout switch **Q5** to turn on, so as to reset the voltage level of the floating diffusion node **Sn** to a corresponding reference level, which is approximately equal to the first reference voltage provided by the first power supply terminal **VDD**. During the reset period **R3**, the voltage levels of the floating diffusion nodes **Sn** of the sensor pixels **20** may be reset line by line in a time-division manner. In detail, during the reset period **R3**, the first switches **Q1** and second switches **Q2** of the sensor pixels **20** arranged in the same line of the pixel array **12** are turned off, and the reset switches **Q3** and the readout switches **Q5** arranged in the same line of the pixel array **12** are turned on, to reset the voltage levels of the floating diffusion nodes **Sn** of the sensor pixels **20** arranged in the same line to corresponding reference levels, which may vary due to manufacturing variations in the sensor pixels **20**.

[0031] During a transferring period **R4** after the reset period **R3**, the first control signal **S1** and the

reset signal S_r are low, and the readout control signal RSL and the second control signal S_2 are high. This causes the first switch Q_1 and the reset switch Q_3 to turn off, and the second switch Q_2 and the readout switch Q_5 to turn on, so as to transfer the voltage level of the first end N_1 of the photodiode PD to the floating diffusion node S_n . During the transferring period R_4 , the voltage levels of the first ends N_1 of the photodiodes PD of the sensor pixels **20** may be transferred line by line in a time-division manner. In detail, during the transferring period R_4 , the first switches Q_1 and the reset switches Q_3 of the sensor pixels **20** arranged in the same pixel line are turned off, and the second switches Q_2 and the readout switches Q_5 of the sensor pixels **20** arranged in the same line are turned on to transfer the voltage levels of the first ends N_1 of the photodiodes PD of the sensor pixels **20** arranged in the same line to the floating diffusion nodes S_n of the sensor pixels **20** arranged in the same line.

[0032] At time point t_c , the second control signal S_2 is pulled from high to low, turning off the second switch Q_2 , and locking the voltage level of the floating diffusion node S_n to a current voltage level of the first end N_1 . Because the photodiode PD starts to accumulate charges at the first end N_1 of the photodiode PD at time point t_b , and the voltage level of the floating diffusion node S_n is locked at time point t_c , the locked voltage level of the floating diffusion node S_n is determined by the total charges accumulated at the first end N_1 of the photodiode PD between time points t_b and t_c . When the image device **1** is in the second mode, the period between t_b and t_c , as labeled **P02**, is the exposure time and directly affects the final voltage level at the floating diffusion node S_n .

[0033] During a readout period R_5 after the transferring period R_4 , the first control signal S_1 , the reset signal S_r , and the second control signal S_2 are low, and the readout control signal RSL is high. This causes the first switch Q_1 , the reset switch Q_3 , and second switch Q_2 to turn off, and the readout switch Q_5 to turn on, so as to enable the readout circuit **22** to output the readout signal S_{out} based on the voltage level of the floating diffusion node S_n . During the readout period R_5 , the sensor pixels **20** of the image device **1** may be read line by line in a time-division manner. In detail, during the readout period R_5 , the first switches Q_1 , the reset switches Q_3 , and second switches Q_2 of the sensor pixels **20** arranged in the same line of the pixel array **12** are turned off, and the readout switches Q_5 arranged in the same line of the pixel array **12** are turned on, to enable the readout circuits **22** of the sensor pixels **20** arranged in the same line to output readout signals S_{out} based on the voltage levels of the floating diffusion nodes S_n of the sensor pixels **20** arranged in the same line of the pixel array **12**.

[0034] Based on the second timing scheme as illustrated in FIG. 4, the controller **10** manipulates the sensor pixels **20** to readout pixel signals in different times (e.g., the sensing period R_2 and the readout period R_5). In addition, because the voltage level of the floating diffusion node S_n is reset only once per frame period T_2 (i.e., the reset signal S_r has only one pulse in each frame period T_2), and the reset period R_3 is after the sensing period R_2 , the voltage level of the floating diffusion node S_n transferred within the transferring period R_4 of the current frame period would be locked until the reset period R_3 of the next frame period. Therefore, the readout signal S_{out} within the sensing period R_2 of the current frame period is generated according the voltage level of the floating diffusion node S_n locked within the previous frame period. In addition, the readout signal S_{out} within the readout period R_5 of the current frame period is generated according the voltage level of the floating diffusion node S_n transferred from the first end N_1 of the photodiode PD within the transferring period R_4 of the current frame period. Assuming that the voltage level of the readout signal S_{out} of a sensor pixel **20**, located in the m .sup.th row and n .sup.th column of the pixel array **12**, read within the sensing period R_2 is $V_a(m, n)$, and the voltage level of the readout signal S_{out} of the same sensor pixel **20** read within the readout period R_5 is $V_b(m, n)$, a difference $D_2(m, n)$ between $V_a(m, n)$ and $V_b(m, n)$ can be represented by the following equation (2):

[00002] $D_2(m, n) = \text{.Math. } V_a(m, n) - V_b(m, n) \text{ .Math.}$ (2)

[0035] When the image device **1** is in the second mode, the controller **10** manipulates the sensor pixels **20** according to the second timing scheme as illustrated in FIG. **4**, configuring the image device **1** to function as a dynamic vision sensor (DVS) for detecting changes in the scene. The image device **1** in the second mode generates a dynamic vision image IMG2 every frame period T2 according to the difference D2 (m, n) of each sensor pixel **20** of the pixel array **12**.

[0036] In an embodiment of the present invention, the controller **10** switches the image device **1** from the second mode to the first mode according to the difference D2 (m, n) of each sensor pixel **20** of the pixel array **12** and a first threshold. Assuming that the first threshold is Dth, and a number of sensor pixels **20**, within every frame period T2, having the difference D2 (m, n) greater than the first threshold Dth is Na, the controller **10** will switch the image device **1** from the second mode to the first mode if the number Na exceeds or equals a second threshold N2, which is a positive integer less than (M×N). If the number Na is less than the second threshold N2, the image device **1** remains in the second mode.

[0037] After controller **10** switches the image device **1** from the second mode to the first mode, the device may remain in the first mode for a predefined period. The predefined period can be, for example, 30 seconds, 1 minute, 3 minutes, or longer. While in the first mode, the image device **1** can generate a series of images (IMG1) of the scene at a specified frame rate (e.g., 25 Hz, 30 Hz, 50 Hz, and 60 Hz). In essence, the image device **1** captures a video stream in the first mode for the predefined period. Once this predefined period elapses, the controller **10** switches the image device **1** back to the second mode to detect changes in the scene again.

[0038] The image device **1** may further comprise a differencing circuit **14** coupled to the pixel array **12** and the controller **10**. When the image device **1** is in the first mode, the differencing circuit **14** calculates the foresaid difference D1 (m, n) of each sensor pixel **20** of the pixel array **12**. When the image device **1** is in the second mode, the differencing circuit **14** calculates the foresaid difference D2 (m, n) of each sensor pixel **20** of the pixel array **12**.

[0039] The image device **1** may further comprise a demultiplexer **16**, an analog-to-digital converting (ADC) circuit **17**, and a comparison circuit **19**. The demultiplexer **16** comprises an input port coupled to the output port of the differencing circuit **14** for receiving the difference D1 (m, n) or D2 (m, n). The demultiplexer **16** further comprises two output ports, one of the output ports is coupled to an input port of the ADC circuit **17**, and the other output port is coupled to an input port of the comparison circuit **19**. When the image device **1** is in the first mode, the controller **10** controls the demultiplexer **16** to output the difference D1 (m, n) of each sensor pixel **20** of the pixel array **12** to the ADC circuit **17**, and the ADC circuit **17** generates the image IMG1 of the scene according to the difference D1 (m, n) of each sensor pixel **20**. When the image device **1** is in the second mode, the controller **10** controls the demultiplexer **16** to output the D2 (m, n) of each sensor pixel **20** to the comparison circuit **19**, and the comparison circuit **19** generates the dynamic vision image IMG2 according to the difference D2 (m, n) of each sensor pixel **20**.

[0040] According to the embodiments of the present invention, the image device **1** uses the same sensor pixels **20** to serve as a global shutter image sensor in the first mode and a dynamic vision sensor (DVS) in the second mode. If the image device **1** in the second mode detects huge changes of the scene, the controller **10** may switch the image device **1** from the second mode to the first mode to generate a video stream by capturing the images of the scene. By using the same sensor pixels **20** in both modes, the hardware resources of the image device **1** are maximized, avoiding the need for separate sensor pixels for each mode. In addition, the ability to switch from the second mode to the first mode to generate a video stream when large changes in the scene are detected increases the flexibility and adaptability of the system, allowing the image device **1** to better handle different scenes and requirements. The image device **1** can remain in the second mode when there are not significant changes in the scene, reducing unnecessary computations and energy consumption, thereby increasing the efficiency of the system. Furthermore, the controller **10** can quickly switch modes to generate a video stream when large changes in the scene are detected,

which helps to improve the real-time performance of the system, enabling the image device 1 to capture and process important scene changes in a timely manner.

[0041] Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. An image device, comprising: a pixel array, comprising a plurality of sensor pixels; and a controller coupled to the pixel array, and configured to switch the image device between a first mode and a second mode; wherein when the image device is in the first mode, the controller manipulates the sensor pixels according to a first timing scheme, configuring the image device to function as a global shutter image sensor for capturing images of a scene; and wherein when the image device is in the second mode, the controller manipulates the sensor pixels according to a second timing scheme, configuring the image device to function as a dynamic vision sensor (DVS) for detecting changes in the scene.
2. The image device of claim 1 further comprising a differencing circuit coupled to the pixel array and the controller, and configured to calculate differences between first readout signals and second readout signals; wherein the image device generates an image of the scene in the first mode or a dynamic vision image in the second mode according to the differences.
3. The image device of claim 2 further comprising: an analog-digital converting (ADC) circuit; a comparison circuit; and a demultiplexer having an input port coupled to an output port of the differencing circuit, a first output port coupled to an input port of the ADC circuit, and a second output port coupled to an input port of the comparison circuit; wherein when the image device is in the first mode, the controller controls the demultiplexer to output the differences to the ADC circuit for generating the image of the scene; and wherein when the image device is in the second mode, the controller controls the demultiplexer to output the differences to the comparison circuit for generating the dynamic vision image.
4. The image device of claim 1, wherein the controller manipulates the sensor pixels to perform correlated double sampling (CDS) operations based on the first timing scheme.
5. The image device of claim 1, wherein the controller manipulates the sensor pixels to readout pixel signals in different times based on the second timing scheme.
6. The image device of claim 1, wherein the each sensor pixel comprises: a photodiode configured to detect light; a first switch coupled between a first power supply terminal and a first end of the photodiode, and configured to reset a voltage level of the first end of the photodiode; a second switch coupled between the first end of the photodiode and a floating diffusion node, and configured to transfer the voltage level of the first end of the photodiode to the floating diffusion node; and a reset switch coupled between the first power supply terminal and the floating diffusion node, and configured to reset a voltage level of the floating diffusion node.
7. The image device of claim 6, wherein the first timing scheme comprises: during a first reset period, turning on first switches of all of the sensor pixels to reset voltage levels of first ends of photodiodes of all of the sensor pixels; during a first sensing period after the first reset period, turning off the first switches of all of the sensor pixels and turning off second switches of all of the sensor pixels to accumulate charges at the first ends of the photodiodes of all of the sensor pixels; and during a first transferring period after the first sensing period, turning off the first switches of all of the sensor pixels, turning off reset switches of all of the sensor pixels, and turning on the second switches of all of the sensor pixels to transfer the voltage levels of the first ends of the photodiodes to floating diffusion nodes of all of the sensor pixels.
8. The image device of claim 7, wherein the first timing scheme further comprises: during the first

reset period, turning on the reset switches of all of the sensor pixels to reset voltage levels of the floating diffusion nodes of all of the sensor pixels.

9. The image device of claim 7, wherein the first timing scheme further comprises: during a second reset period before the first transferring period and after the first sensing period, turning on the reset switches of all of the sensor pixels to reset voltage levels of the floating diffusion nodes of all of the sensor pixels.

10. The image device of claim 7, wherein the each sensor pixel further comprises: a readout circuit coupled to the second switch and the reset switch, and configured to output a readout signal based on the voltage level of the floating diffusion node of the each sensor pixel; wherein the first timing scheme further comprises: during a first readout period after the first transferring period, turning off first switches, second switches, and reset switches of sensor pixels arranged in a same pixel line of the pixel array, and enabling readout circuits of the sensor pixels arranged in the same pixel line to output first readout signals based on voltage levels of floating diffusion nodes of the sensor pixels arranged in the same pixel line; during a third reset period after the first readout period, turning off the second switches of the sensor pixels arranged in the same pixel line and turning on the reset switches of the sensor pixels arranged in the same pixel line to reset the voltage levels of the floating diffusion nodes of the sensor pixels arranged in the same pixel line to reference levels; and during a second readout period after the third reset period, turning off the first switches, the second switches and the reset switches of the sensor pixels arranged in the same pixel line, and enabling the readout circuits of the sensor pixels arranged in the same pixel line to output second readout signals based on the reference levels.

11. The image device of claim 10, wherein the readout circuit comprises: a transistor, comprising: a first end coupled to the first power supply terminal; a second end; and a control end coupled to the floating diffusion node; and a readout switch, configured to control electrical connection between the second end of the transistor and an output end of the readout circuit, wherein the readout signal is outputted via the output end.

12. The image device of claim 6, wherein the second timing scheme comprises: during a first reset period, turning off second switches and reset switches of sensor pixels arranged in a same pixel line of the pixel array, and turning on first switches of the sensor pixels arranged in the same pixel line to reset voltage levels of first ends of photodiodes of the sensor pixels arranged in the same pixel line; during a first sensing period after the first reset period, turning off the first switches and the second switches of the sensor pixels arranged in the same pixel line to accumulate charges at the first ends of the photodiodes of the sensor pixels arranged in the same pixel line, turning off the reset switches of the sensor pixels arranged in the same pixel line, and outputting first readout signals based on voltage levels of floating diffusion nodes of the sensor pixels arranged in the same pixel line; during a second reset period after the first sensing period, turning off the first switches and the second switches of the sensor pixels arranged in the same pixel line of the pixel array, and turning on the reset switches of the sensor pixels arranged in the same pixel line to reset the voltage levels of the first ends of the photodiodes of the sensor pixels arranged in the same pixel line; during a first transferring period after the second reset period, turning off the first switches and the reset switches of the sensor pixels arranged in the same pixel line, and turning on the second switches of the sensor pixels arranged in the same pixel line to transfer the voltage levels of the first ends of the photodiodes of the sensor pixels arranged in the same pixel line to the floating diffusion nodes of the sensor pixels arranged in the same pixel line; and during a first readout period after the first transferring period, turning off the first switches, the second switches and the reset switches of the sensor pixels arranged in the same pixel line, and outputting second readout signals based on the voltage levels of the floating diffusion nodes of the sensor pixels arranged in the same pixel line.

13. The image device of claim 12, wherein the each sensor pixel further comprises: a readout circuit, coupled to the second switch and the reset switch, and configured to output a readout signal

based on the voltage level of the floating diffusion node of the each sensor pixel.

14. The image device of claim 13, wherein the readout circuit comprises: a transistor, comprising: a first end coupled to the first power supply terminal; a second end; and a control end coupled to the floating diffusion node; and a readout switch, configured to control electrical connection between the second end of the transistor and an output end of the readout circuit, wherein the readout signal is outputted via the output end.
